

1

A state can be:

- recurrent: if, starting from it, you go back with prob. 1
- pos. recurrent: if average return time is finite
- null recurrent: if you go back with prob. 1, but average return time is infinite

Assume no positive recurrent states

$$N = |E| < +\infty \quad \text{numb. of states}$$

$$\Rightarrow \sum_{j=1}^N P_{ij}^{(n)} = 1 \quad \forall i \in E, \quad n \geq 0$$

$$\Rightarrow 1 = \lim_{n \rightarrow \infty} \sum_{j=1}^N P_{ij}^{(n)} = \sum_{j=1}^N \lim_{n \rightarrow \infty} P_{ij}^{(n)} = 0$$

$\downarrow$ since $|E|$ is finite $\downarrow$ prob. are positive only for pos. recurrent states

$\Rightarrow$ contradiction

Then, suppose there's one null recurrent state which will then belong to a finite null recur. class.

Since a recurrent class is a MC by itself, this isn't possible

2 Prove that for a renewal process $E[S_{N(t)+1}] = E[X](M(t) + 1)$.

Let $A(t) = E[S_{N(t)+1}] \leftarrow$ first epoch of the first renewal strictly after t
 $X_1 = x$

Idea: calculate mean value of the next renewal time after t

If $x > t \Rightarrow$ no renewal before $t \Rightarrow S_{N(t)+1} = x$

If $x \leq t \Rightarrow$ process renews at time x , restarts independently

Must wait the first renewal after residual time $t-x$:

$$E[S_{N(t)+1} | X_1 = x] = \begin{cases} x & x > t \\ x + A(t-x) & x \leq t \end{cases}$$

Integrating for F we obtain $A(t) = \mu + \int_0^t A(t-x) dF(x)$

This is a renewal equation with constant forcing term μ

Standard solution of

$$Z(t) = a(t) + \int_0^t Z(t-x) dF(x)$$

Here $a(t) = \mu$, so

$$A(t) = \mu + \mu \int_0^t dM(x) = \mu (1 + M(t)) \quad \square$$

Prove that for a Markov chain the n -step transition probabilities, $P_{ij}^{(n)}$, satisfy the relationship

3

$$P_{ij}^{(n)} = \sum_m P_{im}^{(k)} P_{mj}^{(n-k)}, k = 0, 1, \dots, n$$

Starting from state i , decompose the event $\{X_n = j\}$ according to the intermediate state at time k .

$$P_{ij}^{(n)} = P(X_n = j \mid X_0 = i) = \dots$$

$$\dots = \sum_m P(X_k = m \mid X_0 = i) P(X_n = j \mid X_k = m, X_0 = i)$$

By the Markov property, conditional for $X_k = m$, the future after time k is independent of the past

The prob to move from m to j in $n-k$ steps is $P_{mj}^{(n-k)}$

Hence:

$$P_{ij}^{(n)} = \sum_m P_{im}^{(k)} P_{mj}^{(n-k)}$$

4 State and prove the elementary renewal theorem.

If $\mu = E[X_1] \in (0, \infty)$ then

$$\lim_{t \rightarrow \infty} \frac{M(t)}{t} = \frac{1}{\mu}$$

Proof: we know that $A(t) = E[S_{N(t)+1}]$

Lower bound:

Since $S_{N(t)+1} > t$

$$t < \mu(M(t) + 1)$$

so

$$\frac{M(t)}{t} > \frac{1}{\mu} - \frac{1}{t}$$

Therefore:

$$\lim_{t \rightarrow \infty} \frac{M(t)}{t} \geq \frac{1}{\mu}$$

Upper bound: $a > 0$, define truncated interarrival times

$$Y_n = \min(X_n, a) \quad \mu_a = E[Y_1]$$

Let $N_a(t) =$ renewal count

$M_a(t) =$ renewal function

Since $Y_n < X_n \Rightarrow N(t) \leq N_a(t)$ and $M(t) \leq M_a(t)$

Also, $Y_n \leq a$, so:

$$S_{N_a(t)+1}^a \leq t + a$$

$$\Rightarrow \mu_a(M_a(t) + 1) = E[S_{N_a(t)+1}^a] \leq t + a$$

$$\text{Thus: } \frac{M(t)}{t} \leq \frac{M_a(t)}{t} \leq \frac{t+a}{\mu_a t} - \frac{1}{t}$$

$$\Rightarrow \lim_{t \rightarrow \infty} \frac{M(t)}{t} \leq \frac{1}{\mu_a} \rightarrow a \rightarrow \infty \rightarrow \lim_{t \rightarrow \infty} \frac{M(t)}{t} \leq \frac{1}{\mu}$$

Combine limits: $\lim_{t \rightarrow \infty} \frac{M(t)}{t} = \frac{1}{\mu}$

5 Prove that in a Markov chain the period is a class property.

If states i and j communicate, then they have the same period

$$i \leftrightarrow j \Rightarrow d(i) = d(j)$$

Proof.

$S_i = \{s \geq 1 : P_{ii}^{(s)} > 0\} \rightarrow$ set of possible return times to i

$$d(i) = \gcd S_i$$

Since $i \leftrightarrow j$, $\exists m, n \geq 1$ s.t. $P_{ij}^{(m)} > 0$, $P_{ji}^{(n)} > 0$

Take any $s \in S_i$. Then P_i

By joining paths $j \rightarrow i \rightarrow i \rightarrow j$

$$P_{jj}^{(n+s+m)} \geq P_{ji}^{(n)} P_{ii}^{(s)} P_{ij}^{(m)} > 0$$

Thus $n+s+m, n+2s+m \in S_j$, so:

$$d(j) \mid n+s+m \quad d(j) \mid (n+2s+m)$$

Subtract

$$d(j) \mid s$$

Since $\forall s \in S_i$, $d(j) \mid d(i)$, symmetry gives $d(i) \mid d(j)$

Hence, $d(i) = d(j)$

- 6 Prove that for a Poisson process $X(t)$ the statistics of $X(s)$ conditioned on $X(t)$, $s < t$, is binomial, and provide the expression of $P[X(s) = k | X(t) = n]$.

If $(X(t))$ is a Poisson(λ), $0 < s < t$, $0 < k < n$, then

$$P(X(s) = k | X(t) = n) = \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k}$$

Proof: independent increments:

$$X(t) = X(s) + (X(t) - X(s))$$

with $X(s) \sim \text{Pois}(\lambda_s)$ and $X(t) - X(s) \sim \text{Pois}(\lambda(t-s))$ independent. Therefore

$$\begin{aligned} P(X(s) = k | X(t) = n) &= \frac{P(X(s) = k, X(t) - X(s) = n - k)}{P(X(t) = n)} \\ &= \frac{e^{-\lambda_s} \frac{\lambda_s^k}{k!} \cdot e^{-\lambda(t-s)} \frac{(\lambda(t-s))^{n-k}}{(n-k)!}}{e^{-\lambda t} \frac{(\lambda t)^n}{n!}} \\ &= \binom{n}{k} \left(\frac{s}{t}\right)^k \left(1 - \frac{s}{t}\right)^{n-k} \end{aligned}$$

Prove that if states i and j of a Markov chain communicate and i is recurrent, then j is also recurrent.

7

Theorem: If i and j communicate, and i is recurrent, then also j is recurrent

Proof: $i \longleftrightarrow j \Rightarrow \exists m, n \geq 1$:

$$P_{ji}^{(m)} > 0 \quad P_{ij}^{(n)} > 0$$

For $v \geq 0$

$$P_{jj}^{(m+v+n)} \geq P_{ji}^{(m)} P_{ii}^{(v)} P_{ij}^{(n)}$$

Summing over v

$$\sum_{v \geq 0} P_{jj}^{(m+v+n)} \geq P_{ji}^{(m)} P_{ij}^{(n)} \sum_{v \geq 0} P_{ii}^{(v)}$$

Since i recurrent: $\sum_{v \geq 0} P_{ii}^{(v)} = \infty$ by recurrence criterion

Hence:

$$\sum_{l \geq 0} P_{jj}^{(l)} = \infty$$

so j recurrent $\square$

3 For a Poisson process of rate λ , prove that the interarrival times are iid exponential with mean $1/\lambda$.

Let S_n = time between $(n-1)$ st and n th event

$$\textcircled{1} P[S_0 > t] = P[\text{no arrive in } [0, t)) = e^{-\lambda t}$$

$$\Rightarrow S_0 \sim \exp(\lambda) \text{ mean } \frac{1}{\lambda}$$

$$\textcircled{2} P[S_1 > t \mid S_0 = s] = P[\text{no arrivals in } (s, s+t] \mid S_0 = s] \\ = e^{-\lambda t} \Rightarrow S_1 \sim \exp(\lambda) \text{ mean } \frac{1}{\lambda}$$

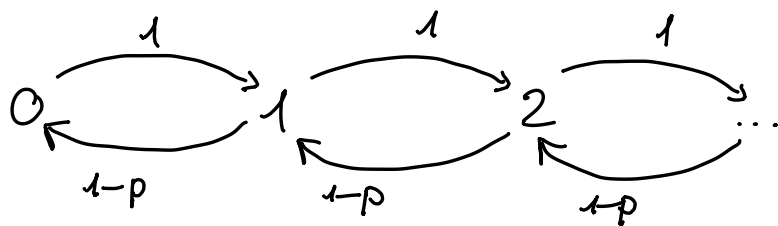
$$\textcircled{3} P[S_n > t \mid S_i = s_i, i=0, \dots, n-1] =$$

$$\Rightarrow P(\text{no arrival in } (s_0 + \dots + s_{n-1}, s_0 + \dots + s_{n-1} + t] \mid S_i = s_i, i=0, \dots, n-1) = \\ e^{-\lambda t} \Rightarrow S_n \sim \exp(\lambda) \text{ with mean } \frac{1}{\lambda}$$

$\downarrow$
ind and stst
increments

□

- 9 Consider a random walk over the non-negative integers with the following transition probabilities: $P_{01} = 1, P_{i,i+1} = p, P_{i,i-1} = q, i > 0$, with $p + q = 1$. Study its behavior, and in particular characterize its recurrence or transiency and derive the steady-state distribution.



$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ \vdots \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & \dots \\ 1-p & 0 & p & 0 & \dots \\ 0 & 1-p & 0 & p & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}$$

Derive the steady state distrib. we need to solve:

$$x_i = \sum_{j=0}^{\infty} x_j P_{ji} = p x_{i-1} + q x_{i+1}$$

where $\sum_{k=0}^{\infty} x_k = 1$ (2)

Firstly:

$$x_0 = q x_1$$

$$x_1 = \frac{x_0}{q}$$

$$x_1 = x_0 + q x_2$$

$$x_2 = \frac{x_1 - x_0}{q} = \frac{x_0 - q x_0}{q^2} = \frac{p x_0}{q^2}$$

$$\Rightarrow \text{generally, } x_i = \frac{1}{p} \left(\frac{p}{q} \right)^i x_0$$

So using (2)

$$1 = \sum_{k=0}^{\infty} x_k = \frac{1}{p} x_0 \sum_{k=0}^{\infty} \left(\frac{p}{q} \right)^k \Rightarrow x_0 = \frac{p}{\sum_{k=0}^{\infty} \left(\frac{p}{q} \right)^k} \quad (3)$$

From 3 we can conclude that:

- If $p < q$ sum converges so chain is positive recurrent
- If $p \geq q$ sum diverges so chain is transient

11 For a Poisson process $X(t)$ of rate λ , state and derive the expression of $P[X(u) = k | X(t) = n]$ for the two cases (i) $0 < u < t, 0 \leq k \leq n$ and (ii) $0 < t < u, 0 \leq n \leq k$.

Binomial Theorem

$$1) 0 < u < t, \quad 0 \leq k \leq n$$

$$= \frac{P[X(u) = k, X(t) = n]}{P[X(t) = n]} =$$

$$= \frac{P[X(u) = k, X(t) - X(u) = n - k]}{P[X(t) = n]} = \dots$$

$$= \binom{n}{k} \left(\frac{u}{t}\right)^k \left(1 - \frac{u}{t}\right)^{n-k}$$

$$2) 0 < t < u \quad 0 \leq n \leq k$$

$$= \frac{P[X(u) = k, X(t) = n]}{P[X(t) = n]} = \frac{P[\overbrace{X(t) = n}^{P[a]}, \overbrace{X(u) - X(t) = k - n}^{P[b]}}]{P[X(t) = n]} =$$

= disjoint intervals $\Rightarrow$ independent =

$$= \frac{\cancel{P[a]} P[b]}{P[\cancel{X(t) = n}]} = \text{stationary increments} =$$

$$= P[X(u-t) = k-n] = \frac{e^{-\lambda(u-t)} (\lambda(u-t))^{k-n}}{(k-n)!}$$

12 For a renewal process, state precisely (also providing a formal proof) what is the value of

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu}$$

$$\lim_{t \rightarrow \infty} \frac{N(t)}{t} = \frac{1}{\mu} \quad \text{w. p. 1}$$

Proof:

$$S_{N(t)+1} \leq t \leq S_{N(t)+1}$$

$$\frac{S_{N(t)}}{N(t)} \leq \frac{t}{N(t)} \leq \frac{S_{N(t)+1}}{N(t)+1}$$

$$\lim_{t \rightarrow \infty} \frac{S_{N(t)}}{N(t)} = \lim_{t \rightarrow \infty} \frac{S_n}{n} = \mu \quad \text{w. p. 1}$$

$$N(t) \rightarrow \infty$$

$$\text{as } t \rightarrow \infty$$

$$\lim_{t \rightarrow \infty} \frac{S_{N(t)+1}}{N(t)+1} = \frac{S_{N(t)+1}}{N(t)+1} \cdot \frac{N(t)+1}{N(t)} = \mu \cdot 1 \quad \text{w. p. 1.}$$

$$\Rightarrow \mu \leq \lim_{t \rightarrow \infty} \frac{t}{N(t)} \leq \mu \quad \text{w. p. 1}$$